**Learning Objective 7.3:** I can use Ampere's Law to calculate the magnetic field at an observation point for simple magnetostatic systems.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Circulation and enclosed current.** Three long straight wires run perpendicular to the page. Wire 1 carries $5$ A out of the page, wire 2 carries $3$ A into the page, and wire 3 carries $2$ A out of the page. A closed path $C$ in the page encloses wires 1 and 2 but not wire 3.

   (a) Find $\oint_C \mathbf{H}\cdot d\mathbf{l}$ when we traverse $C$ counterclockwise.

   circulation = ____

   (b) A second path encloses all three wires and we traverse it clockwise. Find its circulation.

   circulation = ____

   (c) Ampere's law gives the circulation around $C$ exactly. Explain why it does not give the field **H** at a point on $C$ for this arrangement.

2. **A round copper wire.** A long copper wire of radius $a = 1$ mm carries a steady $I = 5$ A, spread uniformly over its cross section.

   (a) Use Ampere's law to derive $H_\phi(\rho)$ inside the wire.

   (b) Find $H_\phi$ at $\rho = 0.5$ mm, $1$ mm, and $2$ mm.

   $0.5$ mm: ____

   $1$ mm: ____

   $2$ mm: ____

   (c) Find $B$ at the surface of the wire.

   $B(a)$ = ____

   (d) Use the point form $\nabla \times \mathbf{H} = \mathbf{J}$ to recover the current density from your answer to part (a), and check it against $I/(\pi a^2)$.

3. **The point form with a nonuniform current.** Inside a long wire of radius $a = 2$ mm, the measured field is $\mathbf{H} = \hat{\phi}\, H_0\, (\rho/a)^2$ with $H_0 = 100$ A/m.

   (a) Find the current density $\mathbf{J}(\rho)$ inside the wire.

   (b) Find the total current in the wire two ways: from Ampere's law on a circle of radius $a$, and by integrating $\mathbf{J}$ over the cross section.

   $I$ = ☐

   (c) Find $J$ at the surface of the wire.

   $J(a)$ = ☐

4. **Two parallel wires.** Two long parallel wires $2$ cm apart carry $10$ A in opposite directions, as in a two-wire line.

   (a) Find $H$ midway between the wires.

   $H$ = ☐

   (b) Find $H$ on the line through both wires, $1$ cm outside one of them.

   $H$ = ☐

**Documentation:** _____